Challenge 1: AI-Native Directory

What you will build

You will build an intuitive discovery experience that moves beyond traditional "what–where" search — one that deeply understands user intent and transforms raw directory data into meaningful, context-aware results.

The goal is not to improve search.
The goal is to rethink how people discover local businesses.

Context

Most directories today work like this:

> User types → gets a list → filters → decides

This model is breaking.

Users think in problems, not keywords. They don't always know what they want. They expect guidance, not lists. And yet the systems they interact with are built entirely around the assumption that they do know — that they can type the right word, pick the right category, apply the right filter.

At the same time, the underlying data is unstructured, inconsistent, and never designed to be understood by anything other than a simple keyword match.

Bridging that gap — between messy raw data and an experience that actually understands people — is the core of this challenge.

Your challenge

This challenge has three distinct layers. Strong solutions will address all three.

1. Make directory data usable
Raw listings are a starting point, not a finished resource. Think about what information is missing, what context would help a system understand a business beyond its name and category, and how existing data can be enriched and structured to support intent-based retrieval.

2. Build the logic that understands intent
Once the data is ready, the harder question is: how does a system interpret what someone actually needs? This is not about autocomplete or filter refinement. It's about moving from query matching to problem understanding — a logic layer that can bridge a vague, context-rich input to a relevant, explainable result.

3. Design a new discovery experience
The interface is not a wrapper around the logic — it is part of the solution. If the UX doesn't change, the experience hasn't changed. Think about what discovery could feel like when the system actually understands the user. This doesn't have to mean a chatbot. It means rethinking the interaction from the ground up.

What we are NOT looking for

- Chatbot wrappers around existing keyword search
- Slightly improved filter UIs
- "AI recommendations" without traceable logic
- Backend-only solutions with no UX consideration

If the experience doesn't feel different — you're missing the point.

Where to start

Think in situations, not categories:

> "I just moved here — what should I try?"
> "Find me a quiet place to work nearby"
> "I need help but I don't know what to search for"
> "I have two hours between meetings — where should I go?"
> "I want something good, not just something close"
> "I need someone reliable for a quick repair — but I don't know who to trust"
> "I want to do something different this weekend"

These are not search queries. They are the real starting points.

Technical hints (optional)

- Semantic search / embeddings
- Intent classification
- Contextual ranking logic
- Lightweight LLM usage

Keep it practical. The best solution is the one that works reliably, not the one with the most moving parts.

Resources & Setup

Available data
We provide a cleaned SMB dataset including:

- Business name, category, location
- Descriptions and services
- Basic metadata (ratings, attributes)

You are encouraged to enrich it, restructure it, and make it useful for your system.

Optional environment

- Frontend mock: simple directory interface with listing overview, detail pages, and space to integrate your experience
- Backend starter: basic API with endpoints for search and retrieval, with optional hooks for further integration

You do NOT have to use this setup. But if you do, it lets you focus on the actual challenge — building a new discovery experience — rather than infrastructure.

Demo expectations

Show clearly:

1. The user situation you are solving for
2. What your discovery experience looks and feels like
3. How intent is understood — and how that drives results
4. Why your results are meaningfully better

We are not looking for scripted demos. Your system should handle different inputs, generate meaningful outputs dynamically, and work beyond a single predefined scenario.

We are not evaluating: *"How good is your demo script?"*
We are evaluating: *"Does your system actually understand and respond?"*

Evaluation criteria

Criterion	Focus
Relevance	Does it solve a real problem?
Innovation	Does it genuinely rethink discovery?
Usability	Is the experience intuitive and meaningful?
Feasibility	Could this realistically work?
Execution	Does it actually work as shown?

One sentence

Turn a static directory into a system that understands people — not queries.